Chromatic Completeness Does Not Determine Geometric Coordinatizability

Karl Svozil [1]

[1] *Institute for Theoretical Physics, TU Wien, Wiedner Hauptstraße 8-10/136, A-1040 Vienna, Austria* [*]

(Dated: July 10, 2026)

We distinguish two properties of context hypergraphs: *chromatic completeness*, the existence of a single globally consistent nondegenerate spectral labeling, and *geometric coordinatizability*, the existence of an injective orthogonal realization by rays. A strong chromatic number larger than the Hilbert-space dimension precludes only the former. It does not, by itself, preclude an orthogonal realization. This is shown by two three-dimensional hypergraphs with the same strong chromatic number. A completed 25-ray Yu–Oh hypergraph has strong chromatic number four and an explicit edge-faithful realization in $\mathbb{R}^3$. Greechie's G_{32} also has strong chromatic number four, as well as a separating and unital set of two-valued states, but admits no injective orthogonal realization in $\mathbb{C}^3$. We give an elementary algebraic proof: its incidence relations force two distinct vertices to represent the same ray. Thus the strong chromatic number does not determine geometric coordinatizability.

I. INTRODUCTION

An n-uniform *context hypergraph* $H = (V, E)$ consists of a vertex set V and a collection E of n-element hyperedges, called *contexts*. It is an abstract incidence structure until a ray realization is specified. In the intended three-dimensional case a context represents a triple of mutually orthogonal rank-one projectors, or equivalently a triple of mutually orthogonal rays.

A *faithful orthogonal representation* (FOR) of an n-uniform hypergraph H in $\mathbb{C}^n$ is an injective assignment

$$i \longmapsto [v_i] \in \mathbb{P}(\mathbb{C}^n)$$

such that the rays assigned to every context are pairwise orthogonal, and hence admit normalized representatives forming an orthonormal basis. Thus

$$\rho(i) \perp \rho(j) \qquad \text{whenever } \{i, j\} \subseteq e \in E.$$

Throughout, *faithful* means injective on rays; it does not additionally require nonincident vertex pairs to be nonorthogonal. When the latter stronger property holds, we say that the realization is *edge-faithful*. The main negative result below uses only injectivity. The existence of orthogonal ray realizations is central to the quantum logic of the Kochen–Specker theorem [1].

Recent work on chromatic contextuality [2, 3] emphasizes that if a hypergraph requires more than n colors, then it cannot support a representation in which every context is assigned the same fixed set of n spectral labels. This observation is correct, but it concerns an additional layer of structure: the global spectral labeling of an already specified incidence geometry. It does not settle whether the incidence structure itself can be coordinatized by rays.

The purpose of this paper is to separate these two issues. Chromatic completeness is a global labeling condition. Geometric coordinatizability is a projective-geometric realization condition. These two conditions are logically independent. The completed Yu–Oh configuration [4] gives a coordinatizable hypergraph with strong chromatic number four in dimension three [3]. Greechie's G_{32} [5] gives a hypergraph with the same strong chromatic number [2], and even a separating and unital set of two-valued states, but no faithful representation in $\mathbb{C}^3$. Thus the chromatic number, the supply of two-valued states, and geometric coordinatizability are three distinct layers of structure.

II. DEFINITIONS

The term "chromatic number" for hypergraphs is not completely uniform in the literature. Throughout this paper we use the following convention.

Definition 1 (Strong chromatic number). *Let $H = (V, E)$ be a hypergraph. A* strong k-coloring *is a map $c : V \to \{1, \dots, k\}$ such that any two vertices occurring in a common context have different colors. Equivalently, for every context $e \in E$, the restriction $c|_e$ is injective. The least such k is denoted by $\chi(H)$.*

For an n-uniform hypergraph, a strong n-coloring assigns all n colors to every context exactly once. This is the coloring notion relevant to nondegenerate spectral assignments.

Definition 2 (Chromatic completeness). *An n-uniform context hypergraph $H = (V, E)$ is* chromatically complete *if there exist a single set*

$$\Sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{R}$$

of n distinct scalars and a map $c : V \to \Sigma$ such that for every context $e \in E$, the restriction $c|_e : e \to \Sigma$ is a bijection.

Thus each context receives the same spectrum, with each spectral value used exactly once.

[*] karl.svozil@tuwien.ac.at

Remark 3 (Chromatic completeness and strong coloring). *For an n-uniform hypergraph, chromatic completeness is exactly strong n-colorability. The n spectral labels in Definition 2 define a strong n-coloring, because every context receives each label exactly once. Conversely, any strong n-coloring may be relabeled by an arbitrary fixed nondegenerate real spectrum $\Sigma = \{\lambda_1, \ldots, \lambda_n\}$, giving precisely the spectral assignment required in Definition 2.*

Consequently, if $\chi(H) > n$, then H has no chromatically complete spectral labeling. At the level of a fixed projective measurement, the numerical eigenvalues of a nondegenerate self-adjoint observable are labels of its rank-one spectral projections: any injective map from its finite spectrum to $\mathbb{R}$ gives another observable with the same projections. Chromatic completeness asks whether these labels can be chosen globally and context-independently. Its failure need not signal a failure of the underlying ray geometry.

III. INDEPENDENCE OF COLORING AND COORDINATIZATION

The following proposition makes the separation explicit. The examples are already available in dimension three.

Proposition 4 (Independence). *Already for $n = 3$, the strong chromatic number does not determine the existence of a faithful orthogonal representation. In particular, there are two 3-uniform hypergraphs with $\chi = 4$, one of which has an edge-faithful representation in $\mathbb{R}^3$ and one of which has no faithful representation in $\mathbb{C}^3$.*

Proof. The completed Yu–Oh hypergraph of Sec. IV has $\chi = 4$ and an explicit edge-faithful representation in $\mathbb{R}^3$. Greechie's G_{32} has $\chi = 4$, as shown in Sec. V B, whereas Theorem 5 proves that it has no faithful orthogonal representation in $\mathbb{C}^3$. □

IV. A COMPLETED 25-RAY YU–OH HYPERGRAPH

The Yu–Oh construction starts from 13 rays in $\mathbb{R}^3$ [4] (for extensions see Refs. [6–8]). In one standard coordinatization these rays are

$$\begin{aligned}
z_1 &= (1,0,0), & z_2 &= (0,1,0), & z_3 &= (0,0,1),\\
y_1^- &= (0,1,-1), & y_2^- &= (1,0,-1), & y_3^- &= (1,-1,0),\\
y_1^+ &= (0,1,1), & y_2^+ &= (1,0,1), & y_3^+ &= (1,1,0),\\
h_0 &= (1,1,1), & h_1 &= (-1,1,1), & h_2 &= (1,-1,1),\\
h_3 &= (1,1,-1). & & & &
\end{aligned}$$

The partial context hypergraph on these 13 rays admits two-valued states. For example, assign value 1 to vertices 3, 6, and 8, and value 0 to all other vertices. Each of the four complete triads then contains exactly one vertex of value 1, and no orthogonal pair has both vertices assigned value 1. The Yu–Oh inequality nevertheless gives a state-independent conflict with noncontextual models reproducing the relevant quantum predictions [4]. This distinction is useful here: the existence of such partial value assignments is separate from both chromatic completeness and ray coordinatizability.

We use a 3-uniform completion of the Yu–Oh orthogonality graph: for each orthogonal pair among the original 13 rays, a third ray is included so that the pair belongs to a full orthogonal triad. The resulting completed hypergraph has 25 vertices and 16 contexts, and is the completion used in the chromatic-contextuality analysis of Ref. [3].

Let $H_{\mathrm{YO}}^+ = (V_{\mathrm{YO}}^+, E_{\mathrm{YO}}^+)$, with $V_{\mathrm{YO}}^+ = \{1, \ldots, 25\}$, have contexts

$$\begin{aligned}
{E_{\mathrm{YO}}}^+ = \{&\{1,2,3\},\{7,8,9\},\{4,5,6\},\{2,8,5\},\\
&\{3,12,25\},\{9,12,16\},\{6,12,17\},\{3,11,14\},\\
&\{7,11,15\},\{4,11,24\},\{9,10,23\},\{1,10,19\},\\
&\{4,10,18\},\{1,13,20\},\{7,13,21\},\{6,13,22\}\}.
\end{aligned} \tag{1}$$

The first four contexts represent the three $\{y_i^+, z_i, y_i^-\}$ triads and the coordinate triad $\{z_1, z_2, z_3\}$. The remaining contexts complete the orthogonal pairs involving h_0, h_1, h_2, h_3.

An explicit coordinatization is obtained by assigning to vertex i the ray spanned by the following unnormalized vector $r_i \in \mathbb{R}^3$:

$$\begin{array}{c|ccccccccccccc} i & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \\ \hline x_i & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & -1 & 1 & 1 \\ y_i & 1 & 0 & 1 & 1 & 0 & -1 & 0 & 1 & 0 & -1 & 1 & 1 & 1 \\ z_i & 1 & 0 & -1 & 0 & 1 & 0 & 1 & 0 & -1 & 1 & 1 & 1 & -1 \end{array} \tag{2}$$

$$\begin{array}{c|cccccccccccc} i & 14 & 15 & 16 & 17 & 18 & 19 & 20 & 21 & 22 & 23 & 24 & 25 \\ \hline x_i & 2 & 1 & 1 & 1 & 1 & 2 & 2 & 1 & 1 & 1 & 1 & 2 \\ y_i & 1 & 2 & -2 & 1 & -1 & 1 & -1 & -2 & 1 & 2 & -1 & -1 \\ z_i & 1 & -1 & 1 & -2 & -2 & -1 & 1 & -1 & 2 & 1 & 2 & -1 \end{array} \tag{3}$$

That is, $r_i = (x_i, y_i, z_i)^T$, and the represented ray is $[r_i]$. The first 13 vertices are labeled as follows:

$$\begin{aligned} &r_1 = y_1^+, \quad r_2 = z_1, \quad r_3 = y_1^-, \\ &r_4 = y_3^+, \quad r_5 = z_3, \quad r_6 = y_3^-, \\ &r_7 = y_2^+, \quad r_8 = z_2, \quad r_9 = y_2^-, \\ &r_{10} = h_2, \quad r_{11} = h_1, \quad r_{12} = h_0, \quad r_{13} = h_3, \end{aligned}$$

where equality means equality of rays. The remaining twelve vertices are completion rays. For instance,

$$r_{14} \parallel r_3 \times r_{11}, \qquad r_{15} \parallel r_7 \times r_{11}, \qquad r_{23} \parallel r_9 \times r_{10},$$

and similarly for the other added vertices.

Direct evaluation gives $r_i \cdot r_j = 0$ precisely for the pairs $\{i, j\}$ contained in a context of E^+_{YO}, and no two vectors in Eq. (2) are proportional. Thus the coordinates give an edge-faithful orthogonal representation of H^+_{YO} in $\mathbb{R}^3$.

The strong chromatic number of this completed hypergraph is four. A strong 3-coloring of H^+_{YO} would restrict to a proper 3-coloring of the Yu–Oh 13-ray orthogonality graph, because every orthogonal pair among the original 13 rays appears in at least one completed triad.

Here is a short self-contained proof that the latter graph is not 3-colorable. Fix the color of h_0 to be 0. Then y_1^-, y_2^-, y_3^- can use only colors 1 and 2. If they all have the same color, the coordinate triad would have to use three distinct colors while avoiding that common color, which is impossible. Otherwise, by permuting indices and exchanging colors 1 and 2, take

$$(c(y_1^-), c(y_2^-), c(y_3^-)) = (1, 1, 2).$$

The four triads force either

$$\begin{aligned} (c(z_1), c(z_2), c(z_3)) &= (0, 2, 1), \\ (c(y_1^+), c(y_2^+), c(y_3^+)) &= (2, 0, 0), \end{aligned}$$

or the same two triples with the first two entries interchanged. In the first case the three neighbours y_2^-, y_1^+, and y_3^+ of h_2 have all three colors; in the second case the three neighbours y_1^-, y_2^+, and y_3^+ of h_1 do. Hence three colors do not suffice.

On the other hand, the following strong 4-coloring shows that four colors do suffice:

$$\begin{array}{c|ccccccccccccc} i & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \\ \hline c(i) & 0 & 2 & 1 & 0 & 3 & 1 & 0 & 1 & 2 & 1 & 2 & 0 & 2 \end{array} \tag{4}$$

$$\begin{array}{c|cccccccccccc} i & 14 & 15 & 16 & 17 & 18 & 19 & 20 & 21 & 22 & 23 & 24 & 25 \\ \hline c(i) & 0 & 1 & 1 & 2 & 2 & 2 & 1 & 1 & 0 & 0 & 1 & 2 \end{array} \tag{5}$$

Each context in Eq. (1) contains three distinct colors. Therefore

$$\chi(H^+_{\mathrm{YO}}) = 4 > 3,$$

although H^+_{YO} has an explicit FOR in $\mathbb{R}^3$. This is the desired example showing that chromatic obstruction is not geometric non-coordinatizability.

V. GREECHIE'S G_{32} HYPERGRAPH

We now turn to the converse phenomenon: a hypergraph with the same strong chromatic number as the completed Yu–Oh example, but with no faithful orthogonal representation in $\mathbb{C}^3$.

A. The incidence structure

Greechie's diagram G_{32} [5, Fig. 6] has vertex set $\{1,\ldots,15\}$ and ten blocks:

$$\begin{aligned}E(G_{32}) = \{&\{1,2,3\},\{3,4,5\},\{5,6,7\},\{7,8,9\},\\ &\{9,10,11\},\{11,12,1\},\{4,10,13\},\\ &\{6,12,14\},\{8,2,15\},\{13,14,15\}\}.\end{aligned} \tag{6}$$

Each block is intended to represent a triad of mutually orthogonal rays. For a two-valued state $s : V \to \{0,1\}$, write $\mathrm{supp}(s) = \{i : s(i) = 1\}$. The following six supports give all two-valued states of G_{32}:

$$\begin{aligned}&\{3,7,10,12,15\}, \quad \{3,6,8,11,13\}, \quad \{2,5,9,12,13\},\\ &\{2,4,7,11,14\}, \quad \{1,5,8,10,14\}, \quad \{1,4,6,9,15\}.\end{aligned}$$

Each support intersects every block in exactly one vertex. Together they are unital and separating, as noted previously in Ref. [2]; nevertheless, the hypergraph is not 3-colorable. Thus it separates the existence of two-valued states from chromatic completeness. The theorem below shows that it also has a geometric obstruction.

B. The strong chromatic number of G_{32}

The hypergraph G_{32} has strong chromatic number four [2]. A strong 4-coloring is

i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$c(i)$	0	1	2	0	1	0	2	0	1	2	3	1	1	2	3.

(7)

A direct check of Eq. (6) shows that each block contains three distinct colors.

To see that three colors do not suffice, form the dual incidence graph: its vertices are the ten blocks of G_{32}, and each atom is an edge joining the two blocks in which it occurs. Since every atom occurs in exactly two blocks, this dual is cubic. Numbering the blocks in Eq. (6) from left to right, the atom-labelled dual obtained directly from the incidence list is the Petersen graph. A strong 3-coloring of the atoms would be a proper 3-edge-coloring of this cubic graph, because the three atoms incident with each block must receive different colors. The Petersen graph has chromatic index four, not three. Hence G_{32} has no strong 3-coloring, and therefore

$$\chi(G_{32}) = 4.$$

This agrees with the previous chromatic analysis of G_{32} in Ref. [2].

C. Non-coordinatizability theorem

Theorem 5. *The hypergraph G_{32} admits no faithful orthogonal representation in $\mathbb{C}^3$.*

Proof. We use the inner-product convention

$$\langle x, y\rangle = x^\dagger y,$$

so the inner product is conjugate-linear in the first argument and linear in the second.

Assume, for contradiction, that a faithful orthogonal representation exists. Since $\{13,14,15\}$ is a block, its three rays form an orthonormal basis. After applying a global unitary transformation, we may fix

$$v_{13} = e_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}, \qquad v_{14} = e_2 = \begin{pmatrix}0\\1\\0\end{pmatrix}, \qquad v_{15} = e_3 = \begin{pmatrix}0\\0\\1\end{pmatrix}.$$

The three blocks meeting this central triangle force three pairs of vectors into coordinate two-planes. From $\{4,10,13\}$ we obtain

$$v_4 = \begin{pmatrix}0\\a\\b\end{pmatrix}, \qquad v_{10} = \begin{pmatrix}0\\-\bar b\\ \bar a\end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

From $\{6,12,14\}$ we obtain

$$v_6 = \begin{pmatrix}c\\0\\d\end{pmatrix}, \qquad v_{12} = \begin{pmatrix}-\bar d\\0\\ \bar c\end{pmatrix}, \qquad |c|^2 + |d|^2 = 1.$$

From $\{8,2,15\}$ we obtain

$$v_2 = \begin{pmatrix}p\\q\\0\end{pmatrix}, \qquad v_8 = \begin{pmatrix}-\bar q\\ \bar p\\0\end{pmatrix}, \qquad |p|^2 + |q|^2 = 1.$$

These parametrizations are general: in each coordinate two-plane, once one normalized ray is chosen, the orthogonal ray is unique up to phase, and the chosen phases above lose no generality at the level of rays.

We shall use the following elementary projection identity. If $\{x,y,z\}$ is an orthonormal basis of $\mathbb{C}^3$, $u \perp x$, and $w \perp z$, then

$$\langle u, w\rangle = \langle u, y\rangle\langle y, w\rangle. \tag{8}$$

Indeed,

$$w = x\langle x, w\rangle + y\langle y, w\rangle + z\langle z, w\rangle.$$

Taking the inner product with u eliminates the first term because $u \perp x$, while the last coefficient vanishes because $w \perp z$. Only the middle term remains, proving Eq. (8).

Apply Eq. (8) to the block $\{1,2,3\}$, written as the orthonormal basis $\{v_3, v_2, v_1\}$. Since $v_4 \perp v_3$ by the block $\{3,4,5\}$ and $v_{12} \perp v_1$ by the block $\{11,12,1\}$, we get

$$\langle v_4, v_{12}\rangle = \langle v_4, v_2\rangle\langle v_2, v_{12}\rangle. \tag{9}$$

With the parametrization above,

$$\langle v_4, v_{12}\rangle = \overline{b}\,\overline{c}, \qquad \langle v_4, v_2\rangle = \overline{a}\,q, \qquad \langle v_2, v_{12}\rangle = -\overline{p}\,\overline{d}.$$

Thus

$$\overline{b}\,\overline{c} = -\overline{a}\,\overline{d}\,q\,\overline{p},$$

and after complex conjugation,

$$bc = -ad\,\overline{q}\,p. \tag{10}$$

Next apply Eq. (8) to the block $\{7, 8, 9\}$, written as the orthonormal basis $\{v_9, v_8, v_7\}$. Since $v_{10} \perp v_9$ by the block $\{9, 10, 11\}$ and $v_6 \perp v_7$ by the block $\{5, 6, 7\}$, we obtain

$$\langle v_{10}, v_6\rangle = \langle v_{10}, v_8\rangle\langle v_8, v_6\rangle. \tag{11}$$

The explicit inner products are

$$\langle v_{10}, v_6\rangle = ad, \qquad \langle v_{10}, v_8\rangle = -b\,\overline{p}, \qquad \langle v_8, v_6\rangle = -qc.$$

In the last equality the first component of v_8 is conjugated once more by $\langle x, y\rangle = x^\dagger y$; hence the factor is q, not $\overline{q}$. Therefore

$$ad = bc\,\overline{p}\,q. \tag{12}$$

Substituting Eq. (10) into Eq. (12) gives

$$ad = (-ad\,\overline{q}\,p)\overline{p}\,q = -ad\,|p|^2|q|^2.$$

Hence

$$ad\big(1 + |p|^2|q|^2\big) = 0.$$

The factor $1 + |p|^2|q|^2$ is strictly positive, so

$$ad = 0.$$

If $a = 0$, then $|b| = 1$ and

$$v_4 = \begin{pmatrix} 0 \\ 0 \\ b \end{pmatrix} = be_3.$$

Thus $[v_4] = [v_{15}]$, contradicting faithfulness. If $d = 0$, then $|c| = 1$ and

$$v_6 = \begin{pmatrix} c \\ 0 \\ 0 \end{pmatrix} = ce_1.$$

Thus $[v_6] = [v_{13}]$, again contradicting faithfulness. Therefore no faithful orthogonal representation of G_{32} in $\mathbb{C}^3$ exists. □

Remark 6. *The proof never uses the chromatic number of G_{32}. The contradiction comes from the propagation of transition amplitudes through the incidence geometry. The two factorizations forced by opposite parts of the outer cycle are algebraically incompatible with distinctness of the central-spoke rays. Thus the obstruction is geometric rather than chromatic.*

VI. CONCLUSION

The existence of a faithful orthogonal representation, the existence of a separating set of two-valued states, and the existence of a globally consistent spectral coloring are distinct questions. The strong chromatic condition $\chi(H) > n$ rules out chromatic completeness in dimension n, because it rules out a global assignment of the same n eigenvalue labels to every context. It does not rule out a faithful orthogonal representation.

The completed Yu–Oh hypergraph makes this explicit: it has $\chi = 4 > 3$ but is edge-faithfully represented by 25 rays in $\mathbb{R}^3$. Greechie's G_{32} has the same strong chromatic number and a separating and unital set of two-valued states, but no faithful representation in $\mathbb{C}^3$. Thus even the common value $\chi = 4$ does not determine coordinatizability. The obstruction in G_{32} is detected by direct geometric analysis, not by coloring or by the mere presence or absence of two-valued states.

Thus chromatic contextuality concerns global consistency of spectral labels, whereas geometric coordinatizability concerns realizability of incidence relations by rays. These two layers of structure should not be conflated.

ACKNOWLEDGMENTS

I thank Mohammad Hadi Shekarriz for valuable comments and suggestions.

The author used OpenAI Codex (GPT-5) for editorial and structural revision of the manuscript. The author directed this use, independently verified the mathematical claims and references, and accepts full responsibility for the final content.

This research was funded in whole or in part by the *Austrian Science Fund (FWF)* [Grant digital object identifier (DOI): 10.55776/PIN5424624]. The author acknowledges TU Wien Bibliothek for financial support through its Open Access Funding Programme.

[1] S. Kochen and E. P. Specker, The problem of hidden variables in quantum mechanics, Journal of Mathematics and Mechanics (now Indiana University Mathematics Journal) **17**, 59 (1967).

[2] M. H. Shekarriz and K. Svozil, Noncontextual coloring of orthogonality hypergraphs, Journal of Mathematical Physics **63**, 032104 (2022), arXiv:2105.08520.

[3] K. Svozil, Chromatic quantum contextuality, Entropy **27**,

387 (2025), arXiv:2501.15261.

[4] S. Yu and C. H. Oh, State-independent proof of Kochen-Specker theorem with 13 rays, Physical Review Letters **108**, 030402 (2012), arXiv:1109.4396.

[5] R. J. Greechie, Orthomodular lattices admitting no states, Journal of Combinatorial Theory. Series A **10**, 119 (1971).

[6] J. Harding and R. Salinas Schmeis, Remarks on orthogonality spaces, International Journal of Theoretical Physics **64**, 211 (2025), arXiv:2505.13871 [math-ph].

[7] A. Cabello, Simplest kochen-specker set, Physical Review Letters **135**, 190203 (2025), arXiv:2508.07335 [quant-ph].

[8] M. Navara and K. Svozil, Construction of Kochen-Specker sets from mutually unbiased bases (2025), arXiv:2509.08636 [quant-ph].